

FINAL REPORT

Course: Applied Linear Algebra for IT

Course Code: 501032

I. REGULATIONS

This report is a replacement for the final written exam in the second semester of the academic year 2023-2024. Students need to adhere to the following regulations:

- Each report is conducted by a group of **one or two students**.
- For each question, students are required to present **detailed calculation steps**, and the final results.
- The report must be submitted in **PDF format**, and the content must be written based on the report/essay format of the Faculty of Information Technology. **In case students do not follow the Faculty's format, they will receive 0 points for the Report.**
- Each group of students must complete the **self-evaluation form** using the following template:

- <https://docs.google.com/spreadsheets/d/1WkFQLo6uvYalRSqT9dbsryC9QFjScKIV/edit?usp=sharing&ouid=104602653034506388459&rtpof=true&sd=true>
- Download the form as an Excel file (.xlsx extension)

- The report must include the following contents:

a. Chapter 1: Solutions (9.0 points)

Write **detailed calculation steps** and the final results of each question (in the section III). In each calculation step, insert the related Background knowledge (theorems, laws, rules, tests...) in case you apply it, for ex.:

- By Gaussian Elimination algorithm, ...

- Unusual cases (such as identical patterns, signs of plagiarism, etc.) will be scheduled for online discussions with the instructor.

II. Submission guideline

- Filenames of the report and self-evaluation files must be the **Student IDs**, for ex.,
 - A group of only one student with student ID 521H1495 will submit a self-evaluation file named **521H1495.xlsx** and a report file named **521H1495.pdf**
 - A group of two students with student IDs 521H1234 and 522H4321 will submit a self-evaluation file named **521H1234_522H4321.xlsx** and a report file named **521H1234_522H4321.pdf**
- Each student in a group submits **two files** to the "**FinalReport**" assignment on Elearning website of the Theory class.

III. Questions

Question 1 (1.0 point). Given the matrix $A = \begin{pmatrix} 1 & 2 & -1 \\ 2 & 2 & 1 \\ 1 & 2 & a \end{pmatrix}$. Find all values of a for which

$$\det(A) = 0.$$

Question 2 (1.0 point). Solve the following system of linear equations by using Gaussian Elimination method.

$$\begin{array}{ll} \text{a)} & \begin{cases} x + 5y - 2z = 4 \\ 3x - y + z = 3 \\ 5x + y - 2z = 4 \end{cases} \\ \text{b)} & \begin{cases} x + 3y - z = 3 \\ x - 2y + 2z = 4 \\ 2x + y + z = 7 \end{cases} \end{array}$$

Question 3 (1.0 point). Let $v_1 = (1; 1; 1)$, $v_2 = (2; -5; 1)$, $v_3 = (3; 0; 5)$. Show that the set $B = \{v_1, v_2, v_3\}$ is a basis of $\mathbb{R}^3$.

Question 4 (2.0 points). Find a matrix P that diagonalizes $A = \begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$.

Question 5 (1.0 point). Let $S = \{v_1 = (2; 4; 3), v_2 = (2; 4; 2), v_3 = (-6; 4; 2)\}$. Find the coordinate vector of $v = (54, 12, 9)$ relative to S .

Question 6 (1.0 point). Use the Gram-Schmidt orthonormalization process to transform the basis $S = \{u_1 = (4; -8; 8), u_2 = (8; 8; 4), u_3 = (8; -4; -8)\}$ for $\mathbb{R}^3$ into an orthonormal basis.

Question 7 (2.0 points). Consider the vector space $\mathbb{R}^3$ with two bases:

$$\varepsilon = \{\varepsilon_1, \varepsilon_2, \varepsilon_3\} \text{ in which } \varepsilon_1 = (1, 0, 0), \varepsilon_2 = (0, 1, 0), \varepsilon_3 = (0, 0, 1)$$

$$\theta = \{\theta_1, \theta_2, \theta_3\} \text{ in which } \theta_1 = (1, 1, 0), \theta_2 = (0, 1, 1), \theta_3 = (1, 0, 1)$$

- a) Find the transition matrix from the basis ε to the basis θ .
- b) Find the transition matrix from the basis θ to the basis ε .